

Disproof of conjecture 00000008422

the λ -spectrum of block designs

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2026-09-13

Verdict: FALSE

We disprove conjecture 00000008422 as stated. Two witnesses are discussed.

- The small case $(v, k) = (6, 3)$ shows that the criterion “ $\lambda \binom{v}{t} / \binom{k}{t}$ is an integer with $t = k - 1$, and $\lambda \leq \binom{v-2}{k-2}$ ” is *not sufficient*: it predicts $\lambda \in \{1, 2, 3, 4\}$, whereas the attainable set is $\{2, 4\}$. However, this produces only 2 exceptions, and $2 \leq k^2 = 9$. Since the conjecture itself allows “a finite explicit exceptional set with at most k^2 exceptions,” the $(6, 3)$ witness is *absorbed* by that clause: it refutes only the strict “if and only if / the spectrum is complete” reading, not the exception-clause reading.
- The case $(v, k) = (22, 3)$ is decisive. The criterion predicts *every* $\lambda \in \{1, \dots, \binom{20}{1}\} = \{1, \dots, 20\}$, but the necessary replication condition $r = \lambda(v - 1)/(k - 1) = 21\lambda/2$ forces λ to be even. The ten odd values in $\{1, \dots, 20\}$ are therefore unattainable although predicted, and $10 > k^2 = 9$. This violates the exception clause itself, so the conjecture fails under *both* readings.

1 The conjecture, quoted

Definition 1 (λ -spectrum). For given (v, k) , the λ -*spectrum* is the set of realizable λ for which a 2 -(v, k, λ) design exists.

Quoted verbatim from `conjectures/00000008422.md`:

Definition: The lambda spectrum of block designs: the set of realizable lambda values for given (v,k). **Conjecture:** lambda is attainable if and only if lambda $\binom{v}{t} / \binom{k}{t}$ is an integer (with $t = k - 1$) and lambda is at most $\binom{v-2}{k-2}$, so the spectrum is complete; the complement of the spectrum is a finite explicit exceptional set with at most k^2 exceptions. (complete characterization of design lambda spectrum)

Thus the criterion set is

$$\mathcal{C}(v, k) = \left\{ \lambda \in \mathbb{Z}_{>0} : \lambda \frac{\binom{v}{t}}{\binom{k}{t}} \in \mathbb{Z}, t = k - 1, \lambda \leq \binom{v-2}{k-2} \right\}, \quad (1)$$

and the conjecture asserts, up to a “finite exceptional set with at most k^2 exceptions,” that the spectrum equals $\mathcal{C}(v, k)$. We take the exception clause at face value: the number of criterion-predicted values that are *not* attainable is at most k^2 . We show this fails at $(22, 3)$. The $(6, 3)$ case is included first because it is the smallest and because it shows the mechanism.

Throughout, a $2-(v, k, \lambda)$ design has block count and replication number

$$b = \frac{\lambda v(v-1)}{k(k-1)}, \quad r = \frac{\lambda(v-1)}{k-1}. \quad (2)$$

2 The divisibility argument

Lemma 2 (replication identity). *In a $2-(v, k, \lambda)$ design, for every point p the number r of blocks through p satisfies*

$$(k-1)r = (v-1)\lambda. \quad (3)$$

Consequently $(k-1) \mid (v-1)\lambda$, and $2-r = (v-1)\lambda$ when $k = 3$.

Proof. Count the flags “(block B through p , a second point $q \in B \setminus \{p\}$)” in two ways.

By block. There are r blocks through p , and each contains exactly $k-1$ further points, giving $(k-1)r$ flags.

By pair. Group by q . For each of the $v-1$ points $q \neq p$, the pair $\{p, q\}$ lies in exactly λ blocks, so it contributes λ flags, for a total of $(v-1)\lambda$.

Both counts count the same set, so $(k-1)r = (v-1)\lambda$; since r is an integer, $(k-1) \mid (v-1)\lambda$. $\square$

Corollary 3. *For $k = 3$: if $2 \nmid (v-1)$, then λ is even. In particular $(6, 3)$ forces λ even, and $(22, 3)$ forces λ even.*

Proof. $2r = (v-1)\lambda$ with $v-1$ odd gives $2 \mid \lambda$ (as 2 is prime and does not divide $v-1$). $\square$

3 The small witness $(v, k) = (6, 3)$

For $(6, 3)$ we have $t = k-1 = 2$ and

$$\frac{\binom{6}{2}}{\binom{3}{2}} = \frac{15}{3} = 5, \quad \binom{4}{1} = 4.$$

Hence $\lambda \cdot 5$ is an integer for *every* λ , and the bound is $\lambda \leq 4$. The criterion (1) therefore predicts

$$\mathcal{C}(6, 3) = \{1, 2, 3, 4\}.$$

By Lemma 2 a design needs $r = 5\lambda/2 \in \mathbb{Z}$, i.e. λ even, so a design can exist only for $\lambda \in \{2, 4\}$. Both do exist.

Theorem 4. *The attainable set at $(v, k) = (6, 3)$ is exactly $\{2, 4\}$; in particular*

$$\mathcal{C}(6, 3) = \{1, 2, 3, 4\} \neq \{2, 4\}.$$

So the criterion is not sufficient, and the “spectrum is complete” statement is false at $(6, 3)$.

Proof. The divisibility argument excludes $\lambda = 1$ and $\lambda = 3$. For the two remaining values we exhibit designs (a block $\{a, b, c\}$ is written abc).

$\lambda = 2$. The ten triples

$$\begin{array}{c} 012, 013, 024, 035, 045, \\ 125, 134, 145, 234, 235 \end{array}$$

cover every one of the $\binom{6}{2} = 15$ pairs exactly twice; this is verified by direct enumeration in `reproduce.py`. The block count is $b = 2 \cdot 6 \cdot 5 / (3 \cdot 2) = 10$, as required.

$\lambda = 4$. The complete design, consisting of all $\binom{6}{3} = 20$ triples, has each pair in $\binom{4}{1} = 4$ triples; the block count is $b = 4 \cdot 6 \cdot 5 / (3 \cdot 2) = 20$.

Exhaustive confirmation. An exhaustive search over all $\binom{20}{5} = 15504$ five-element sublists of the twenty triples finds *no* 2-(6, 3, 1) design, and over all $\binom{20}{15} = 15504$ fifteen-element sublists finds *no* 2-(6, 3, 3) design. Equivalently, a 2-(6, 3, 3) design is the complement (inside the twenty triples) of a 2-(6, 3, 1) design, because the complete design has $\lambda = 4$; the same enumeration rules it out. The search is carried out in `reproduce.py` and independently in Lean (`no_design_1`, `no_design_3`). $\square$

Remark 5 (why (6, 3) alone is not decisive). The criterion predicts four values and two are unattainable: exactly 2 exceptions. The conjecture allows up to $k^2 = 9$ exceptions, so at (6, 3) the data are *consistent* with the exception-clause reading. A decisive witness must therefore produce more than k^2 exceptions. This is what (22, 3) does.

4 The decisive witness $(v, k) = (22, 3)$

For (22, 3) we again have $t = k - 1 = 2$ and

$$\frac{\binom{22}{2}}{\binom{3}{2}} = \frac{231}{3} = 77, \quad \binom{20}{1} = 20.$$

Hence $\lambda \cdot 77$ is an integer for every λ , and the bound is $\lambda \leq 20$. The criterion predicts all of

$$\mathcal{C}(22, 3) = \{1, 2, 3, \dots, 20\}, \quad |\mathcal{C}(22, 3)| = 20.$$

By Lemma 2, a 2-(22, 3, λ) design requires

$$2r = 21\lambda,$$

so λ must be even (Corollary 3, since 21 is odd).

Theorem 6. *At $(v, k) = (22, 3)$, the ten odd values*

$$1, 3, 5, 7, 9, 11, 13, 15, 17, 19$$

satisfy the criterion (they lie in $\mathcal{C}(22, 3)$) but are not attainable. Consequently the number of criterion-predicted but unattainable values is at least 10, while the conjecture's exception budget is $k^2 = 3^2 = 9$. Therefore

$$10 > 9,$$

and the exception clause itself fails. Conjecture 00000008422 is false under both the strict “if and only if” reading and the “at most k^2 exceptions” reading.

Proof. Each odd λ fails the necessary condition $2r = 21\lambda$: the right-hand side is odd while the left-hand side is even, so no integer replication number r exists and no design (simple or with repeated blocks) can exist. Since all twenty values in $\mathcal{C}(22, 3)$ are predicted and ten of them are odd, the number of exceptions is at least $10 > 9 = k^2$. No enumeration of designs is needed: non-attainability is forced by a necessary condition, so the count is a lower bound that holds whatever the full spectrum is. $\square$

Remark 7. The same phenomenon recurs for every even v with $v \equiv 0$ or $2 \pmod{3}$ large enough that the criterion admits more than k^2 odd values. Already $(24, 3)$ gives $\binom{24}{2}/\binom{3}{2} = 276/3 = 92$ and bound $\binom{22}{1} = 22$, with $11 > 9$ odd values forced out. The smallest decisive case is $(22, 3)$.

5 Numerical summary

(v, k)	t	$\binom{v}{t}$	$\binom{k}{t}$	ratio	bound	$\binom{v-2}{k-2}$	$ \mathcal{C} $	forced out
$(6, 3)$	2	15	3	5		4	4	2
$(7, 3)$	2	21	3	7		5	5	0
$(8, 3)$	2	28	3	9		6	2	1
$(10, 3)$	2	45	3	15		8	8	4
$(22, 3)$	2	231	3	77		20	20	10 > 9
$(24, 3)$	2	276	3	92		22	22	11 > 9

“Forced out” counts the values in $\mathcal{C}(v, k)$ that violate the necessary condition $(k - 1) \mid \lambda(v - 1)$; these are certainly unattainable, so it is a lower bound on the true number of exceptions. At $(22, 3)$ and $(24, 3)$ it already exceeds $k^2 = 9$.

6 Reproducibility

`reproduce.py` (Python 3 standard library only) performs the exhaustive $(6, 3)$ search, exhibits the $\lambda = 2$ and $\lambda = 4$ designs, checks the divisibility argument, and computes the criterion table above. It prints **PASS** and exits 0 exactly when every check holds:

```
$ python3 reproduce.py
...
PASS: all checks verified.
The conjecture 00000008422 is FALSE.
```

7 Formalisation

The accompanying Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`) formalises the $(6, 3)$ computations and the arithmetic kernel of the divisibility argument:

```
theorem design2_is : isDesign design2 2 = true
theorem design4_is : isDesign triples 4 = true
theorem no_design_1 :
  (sublistsLen 5 triples).all (fun c => !(isDesign c 1)) = true
theorem no_design_3 :
  (sublistsLen 5 triples).all
    (fun c => !(isDesign (complement c) 3)) = true
```

```

theorem parity_22_3 (r lam : Nat) (h : 2 * r = 21 * lam) : 2 | lam
theorem decisive_22_3 : ... 3 ^2 < 10

```

The general counting identity (3) of Lemma 2 is proved above on paper; Lean keeps it as the explicit hypothesis of `parity_6_3` and `parity_22_3`. The exhaustive $(6, 3)$ search is run in the kernel by `decide` over all $\binom{20}{5} = 15504$ candidates. See `lean4/README.md` for the full statement table and scope note.

References

- [1] D. R. Stinson, *Combinatorial Designs: Constructions and Analysis*, Springer, 2004. (Existence criteria (2), the replication identity (3), and the nonexistence of a Steiner triple system on 6 points, since $\text{STS}(v)$ exists iff $v \equiv 1, 3 \pmod{6}$.)